

From Collective Object Transportation to Embodied Coordination: Distributed Cascade Force Control with Minimal Information for Soft-Tactile Multi-Robot Pushing

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Abstract—We propose a novel perspective on collective mobility in multi-robot systems, where a shared object is not only transported but also serves as a physical medium for maintaining coordination among robots capable only of pushing. Each robot is equipped with a unique 360° soft-tactile sensor that estimates the contact force location and magnitude with respect to the object. Through our proposed distributed cascade control strategy with a Lyapunov-based stability guarantee, each robot maintains local pushing contact, becoming implicitly coupled as a loosely connected team moving toward a desired direction. This approach is particularly suitable for unstructured outdoor environments where inter-robot communication, sensing, and global information are limited or unreliable, and object shapes vary. We validate the proposed framework through both simulations and hardware experiments under varying object mass properties and external interferences. The results demonstrate robustness and flexibility in maintaining coordinated motion and achieving object transportation across different object shapes, robot positions, and team sizes. This work highlights its potential for deployment in the wild, across diverse field robotics scenarios.

Index Terms—Distributed control, Lyapunov-based stability, Multi-robot system, Pushing force control, Vision-based tactile sensors

I. INTRODUCTION

Multi robot systems are often categorized into two primary domains: multi robot coordination and multi robot object transportation. While multi robot coordination has shown success in structured environments, it faces significant limitations in field robotics applications. This is due to its strong reliance on inter robot communication and relative sensing to maintain formation or distributed planning. In extreme or unstructured environments such as space, deep sea, or post disaster zones, these assumptions often break down. Communication becomes unreliable or unavailable, and sensing the relative positions of teammates may be infeasible due to occlusion, harsh lighting, or lack of infrastructure. These challenges call for more embodied and contact based strategies where coordination emerges from physical interaction rather than explicit messaging.

This leads to the development of strategies that use interaction with the object as a physical medium for coordination. This concept is similar to how ants cooperatively carry food

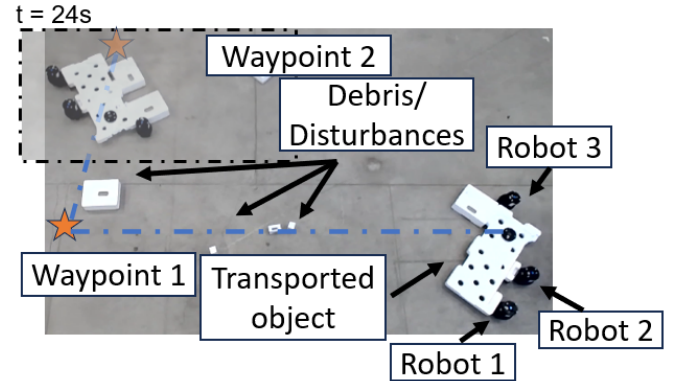


Fig. 1: Three robots push an object or debris toward Waypoint 1, then switch to Waypoint 2 when nearby. Each robot controls its movement and pushing force based on unconventional 360-degree local tactile sensing without knowledge of the object’s state, other robots’ positions, or their applied forces. This interaction with the shared object indirectly enables the robot team to stay coordinated without requiring relative position information.

back to the nest. In such biological systems, the object itself becomes a shared medium through which agents coordinate, without any need for explicit communication.

In multi-robot object transportation, grasp-based teams use fixed connections, so robot positions are naturally constrained relative to the object. In contrast, with pushing, each robot can join or leave the interaction, reposition itself, or remain attached to the object without significantly affecting its movement. However, coordinating multiple pushing robots to transport a rigid object using only local sensing while maintaining stable pushing contact remains a key challenge in the field, as illustrated in Fig.1. Existing state-of-the-art studies typically fall into one of three directions: (a) single mobile manipulators with force sensors at the end effector [1], [2] (b) decentralized push using robot bodies but without force, control [3]–[9], often requiring explicit knowledge of the states of other robots to maintain contact; (c) decentralized

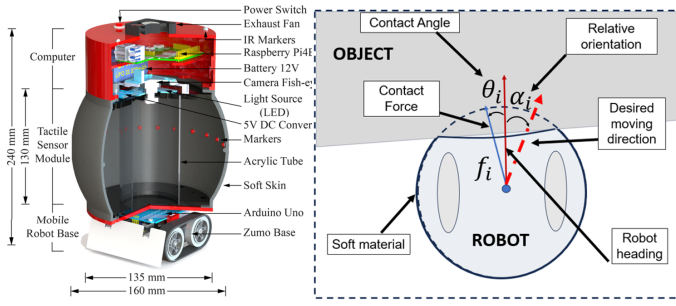


Fig. 2: 360° vision-based soft tactile robot enables real-time estimation of multiple contact force angles and magnitudes using an embedded fisheye camera and a machine learning algorithm.

collaborative manipulation using grasping or caging mechanisms [10]–[14], which require physical fixed contact with the object.

A key challenge arises when a rigid object both translates and rotates in two-dimensional environments. Conventional differential drive robots struggle to maintain pushing contact and apply the desired force simultaneously, especially when the force direction is not aligned with the robot's direction. Our unconventional robot system addresses this challenge by leveraging local 360-degree tactile sensing in combination with distributed cascade pushing force control to enable successful object transportation.

With this distributed control strategy, each robot follows the same force and motion policy, and emergent team behavior leads to coordinated object transportation. This approach is particularly well suited for unstructured or post-disaster environments, where debris and obstacles have irregular shapes and sizes. It takes advantage of the collective strength of multiple small robots and enhances the resilience of the system in the event of individual robot failures. The method is also applicable in scenarios with limited communication or no access to shared global states, making it ideal for unconventional robotic systems operating in real-world conditions.

This work further explores the synergy between distributed pushing force control and emergent coordination, enabled by a novel 360-degree soft tactile sensing sensor. This tactile embodiment helps each robot utilize local information from its interaction with the object, capturing the object's rotational behavior through full-body circular sensing. It allows robots to maintain contact and estimate the resulting torque without requiring knowledge of the object's state or the actions of other robots. This forms the foundation for robust object transport in real-world unstructured environments, even under strict communication and sensing constraints.

II. BACKGROUND

A. Rigid-body object dynamics and Pushing Force Modelling

In this paper, we consider a rigid-body object with mass m , moment of inertia I under the pushing forces $\mathbf{f}_i$ from n agents as described in Figure 3. The object slides across a

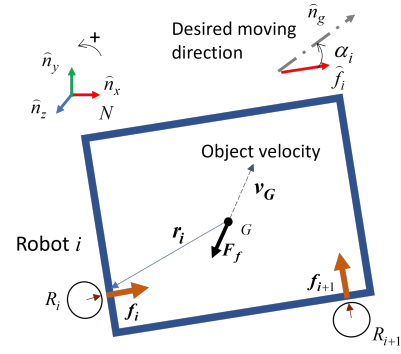


Fig. 3: Free-body diagram of a body/object under a pushing force $\mathbf{f}_i$.

planar frictional environment, and its motion can be decoupled into translation and rotation around the center of mass G . $\mathbf{r}_i$ is the distance from G to robot i . Denote the object's linear velocity as $\dot{\mathbf{v}}_G(t) \in \mathbb{R}^2$ and the angular velocity as $\omega(t)$. These behaviour are described respectively by Newton's law of motion and Euler's rigid body equation:

$$m\dot{\mathbf{v}}_G = \sum_{i=1}^n \mathbf{f}_i + \mathbf{F}_f, \quad (1)$$

$$I\dot{\omega} = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{f}_i) + \mathbf{M}_f, \quad (2)$$

where $\times$ is the cross-product operator. $\mathbf{F}_f$ is the frictional force, $\mathbf{M}_f$ is the frictional torque between the object and surface.

In the scope of this paper, we model the friction between an object and the ground as follows. In order to translate the object, the net robot force magnitude has to be greater than the critical static friction F_{crit} :

$$\left\| \sum_{i=1}^{n_0} \mathbf{f}_i \right\| > F_{crit} \quad (3)$$

Similarly, the object starts rotating if the resultant moment from robots is greater than the maximum static frictional torque M_{crit} :

$$\left\| \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{f}_i) \right\| > M_{crit} \quad (4)$$

When the object is in motion, the friction force and torque can be presented as the zero and first order of motion, respectively:

$$\mathbf{F}_f = -\mu_{0,t} m \frac{\mathbf{v}_G}{\|\mathbf{v}_G\|} - \mu_{1,t} \mathbf{v}_G, \quad (5)$$

$$\mathbf{M}_f = -\mu_{0,r} I \frac{\omega}{\|\omega\|} - \mu_{1,r} \omega, \quad (6)$$

where $\mu_{0,t}$, $\mu_{1,t}$, $\mu_{0,r}$, and $\mu_{1,r}$ are the friction parameters.

Moreover, in a rigid body, the relation of the velocity of a specific point measured by robot i and the object velocity at the center of mass G is described as:

$$\mathbf{v}_i = \mathbf{v}_G + \mathbf{r}_i \times \omega \quad (7)$$

B. Robot System Overview

To address the above challenges, we introduced a 360-degree vision-based soft tactile sensor mounted on top of a differential-drive chassis in our previous work [15], as shown in Figure 2. With calibrated markers and onboard fisheye vision, each robot estimates local contact position and force in real-time. The sensing system achieves mean absolute errors of 1.12 degrees in contact angle and 0.12 newtons in force, as reported in [1]. This platform allows robots to independently explore the object's boundary, adjust their contact points, and initiate pushing without any shared information. Our prior work [16] addressed parallel pushing with fixed formations. In contrast, our approach supports non-parallel configurations and dynamic positioning.

III. PROBLEM FORMULATION

The problem is described as n robots pushing an unknown rigid-body object under frictional planar described in the Background section. During transportation, each robot, namely i , is equipped with a 360-degree tactile sensor described in the Background section to measure pushing force magnitude f_i and contact angle $\theta_{i/r}$ in the robot i 's frame. Moreover, each robot knows task-related-information measured in robot frame: relative orientation $\alpha_{i/r} \in [-\pi, \pi]$ between the desired direction $\hat{n}_g$ and its heading $\hat{r}_{x,i}$. In practice, local sensors, such as a digital compass and the pre-set desired direction for each robot, can obtain such information.

The desired direction $\hat{n}_g$ situated in a world frame N with axis as To transport an object to the , we want the object velocity direction converge to the desired direction:

$$\lim_{t \rightarrow \infty} \arg(\dot{\mathbf{v}}_G(t)) = \theta_{g,N}. \quad (8)$$

Objective 1 (Following Desired Direction) Ensure that the object is transported following the desired direction $\hat{n}_g$ or object motion direction $\arg(\dot{\mathbf{v}}_G(t))$ converge to the desired direction θ_g :

Objective 2 (Object Stabilization) Ensure that the object's angular rate ω is required to converge asymptotically at 0 to stabilize rotation motion and help to push robots to maintain pushing contacts:

$$\lim_{t \rightarrow \infty} \omega(t) = 0. \quad (9)$$

The robots do not know or are impractical to measure the following parameters in complex environments:

- object's mass properties, including mass m , inertia mass I , and the position of the rotation axis G or the center of mass.
- frictional coefficients $\mu_{0,t}$, $\mu_{1,t}$, $\mu_{0,r}$, and $\mu_{1,r}$.
- relative distance and relative bearing to other robots.
- other robot information during transportation since there is no communication among agents to exchange the information that they have.

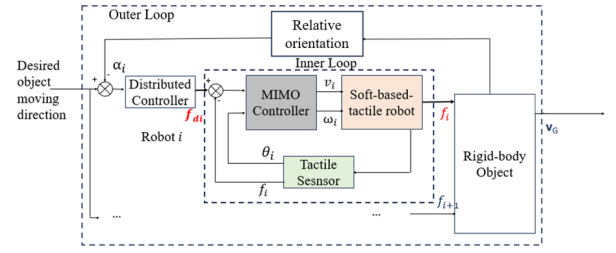


Fig. 4: Distributed cascade force control relies on local information with inner and outer loop.

IV. CONTROL FRAMEWORK

A. Pre-transportation Phase

Robots are split into two pushing groups based on their relative orientation to the desired goal direction. The remaining robots maintain only a small contact force with the object without actively contributing to its motion. The definitions of the robot groups are as follows:

The *Right group* comprises n^R robots that satisfy the condition:

$$\text{Right group} = \left\{ j \mid \alpha_j - \theta_j \in \left[0, \frac{\pi}{2}\right], \forall j \in [1, n^R] \right\} \quad (10)$$

Similarly, the *Left group* includes robots that satisfy:

$$\text{Left group} = \left\{ k \mid \alpha_k - \theta_k \in \left[-\frac{\pi}{2}, 0\right], \forall k \in [1, n^L] \right\} \quad (11)$$

Note that the number of robots in each group, n^R and n^L , is not known in advance.

In the pre-transportation phase, each robot group performs trial pushing around the object's boundary to: (i) overcome static friction by gradually increasing each robot's force until the object starts translating. The pushing force of each robot at this moment is denoted as f_{j0}^L for robot j in the *Left group*, and f_{k0}^R for robot k in the *Right group*; and (ii) estimate the sign of the produced moment using the rate of change of the contact angle from the robot's tactile sensor:

$$\text{sign}(\dot{\theta}_{i/r}(t)) = \text{sign}(\omega) = \text{sign}\left(\sum_{i=1}^{n_p} (f_i d_i)\right) \quad (12)$$

For each pushing position, the sign of the produced moment is fixed during transportation and can be positive, negative, or zero. To minimize the object's angular rate as stated in Problem 2, at least two robot groups must satisfy one of the following configurations:

Scenario 1: Both groups produce negligible moments (pure translation) when all lever arms are approximately zero:

$$d_j^R \approx 0, \quad d_k^L \approx 0 \quad (13)$$

Scenario 2: The *Left group* produces a negative moment, and the *Right group* produces a positive moment, i.e.:

$$d_j^R > 0 \quad \forall j \in [1, n^R], \quad d_k^L < 0 \quad \forall k \in [1, n^L] \quad (14)$$

These trial pushes can continue until one of the above scenarios is satisfied. This can be achieved by: (i) having

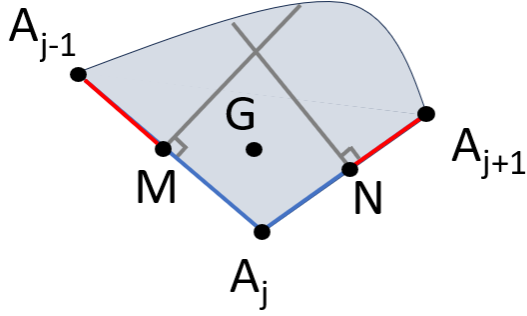


Fig. 5: A part of a convex polygon-shaped object with the vertex A_j closest to the center of mass G . There is always at least one pushing position where robots are distributed on $A_{\text{mod}(j-1,n)}M$ and $NA_{\text{mod}(j+1,n)}$

a sufficient number of robots distributed around the object boundary, (ii) allowing robot groups to push randomly or follow a predefined policy, and (iii) letting the object rotate as a result of the trial pushes.

B. Feasible pushing conditions for convex polygon shape

Given the above conditions, there are feasible pushing positions for objects with the most common shapes: convex polygons, circles, and ellipses.

Considering an object with a cross-section in the shape of a convex polygon with n vertices from A_0 to A_{n-1} (counter-clockwise), and unevenly distributed weight with the center of mass at G as in the Fig. 5. Without loss of generality, let A_j be the vertex closest to G , M and N respectively the midpoints of $A_{\text{mod}(j-1,n)}, A_j$, and $A_j, A_{\text{mod}(j+1,n)}$, where $\text{mod}(a, b)$ represents the modulo operation which yields the remainder of the division of a by b . Then, there always exists a suitable pushing position for the robots to push a convex polygon-shaped object, determined by: $A_{\text{mod}(j-1,n)}M$ and $NA_{\text{mod}(j+1,n)}$.

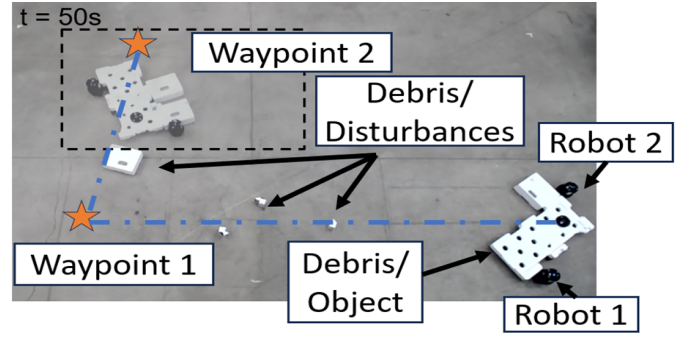
C. Transportation phase with Distributed cascade force control

The control framework is structured as a distributed cascade system with two tightly coupled layers, as shown in Figure 4. The outer loop plays a high-level role in the distributed control policy for robots in each group:

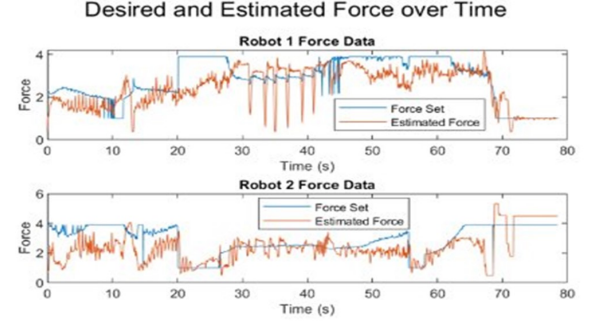
$$f_j^R(t) = f_{j0}^L \left(\frac{k_0}{\sin(|\alpha_i(t) - \theta_{i/r}(t)|)} - k_1 \omega(t) \right), \quad (15)$$

$$f_k^L(t) = f_{k0}^R \left(\frac{k_0}{\sin(|\alpha_i(t) - \theta_{i/r}(t)|)} + k_1 \omega(t) \right). \quad (16)$$

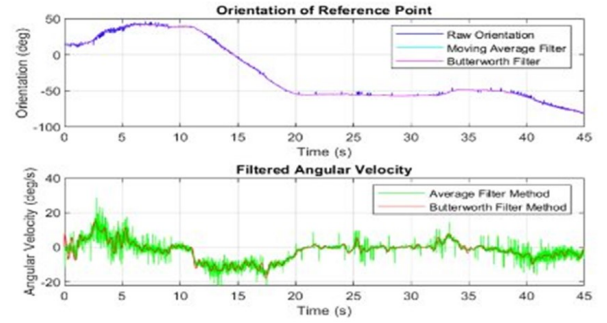
This ensures that the resultant force points toward the goal and that torques from different robot groups cancel out. Lyapunov stability analysis, combined with the invariant set theorem, guarantees asymptotic convergence of angular velocity and



(a) Robot system before and after 50 seconds of video playback (equivalent to 150 seconds in real-time due to 3x speed)



(b) Desired and estimated pushing forces measured from two robots.



(c) Object orientation changes as the box moves toward new waypoints. The object's angular velocity converges to zero.

Fig. 6: Experiment results of the scenario with two robots.

orientation. The gain k_0 adjusts the resultant force magnitude that helps overcome static friction or increase translational speed for the object. A larger k_1 results in a faster decay rate of the Lyapunov function, affecting the object's rotation behavior.

V. EXPERIMENTS AND RESULTS

We validate the proposed system in MATLAB simulation and real-world experiments using two and three robots pushing an object toward a desired goal direction. The experimental results for the case of two robots are described in Figure 6. A collection of video demonstrations from the hardware experiments is available online.¹

¹<https://drive.google.com/drive/folders/1JR2ICxj5mS4Pj4K5xCLtVbvRF7-217D-?usp=sharing>

VI. CONCLUSION

The proposed system demonstrates that minimal-information multi-robot object transportation is feasible using only local tactile sensing and physical interaction with the object. The use of the shared object as a coordination medium allows robots to implicitly maintain formation and apply coordinated forces without any explicit communication or relative positioning.

A key advantage of this approach is its scalability. Since each robot follows the same local policy and requires no global state, the system can be extended to larger teams or deployed in unstructured environments such as collapsed buildings, underwater, or planetary terrains. However, there are limitations. The current method assumes that each robot can detect the resulting rotation of the object during the pre-trial pushing phase. While this can be achieved using onboard sensing or object-mounted markers, such capabilities may be harder to maintain in rough outdoor conditions. Additionally, this approach assumes the object is sufficiently rigid and does not deform significantly during interaction. Future work will explore methods for estimating object response more robustly under noise, integrating this framework with adaptive or learning-based tactile models, and scaling to larger robot teams with heterogeneous capabilities. The idea of object-mediated coordination also opens opportunities for decentralized planning and adaptive team formation based purely on physical interactions.

VII. APPENDIX-MATHEMATICAL PROOF OF STABILITY AND CONVERGENCE

A. Stability and convergence of LDFC

The robot coefficients f_{j0}^L and f_{k0}^R are chosen such that the net pushing force of each group approximates the static friction threshold:

$$\sum_{j=1}^{n^R} f_{j0}^L \approx \sum_{k=1}^{n^L} f_{k0}^R \geq F_{\text{crit}}. \quad (17)$$

There are two proofs for two theorems (Theorem 1 and 2) corresponding to two stated problems (Problem 1 and Problem 2), respectively.

Theorem 1. If each robot in the described system follows the proposed control law (as in eqs.15 and 16), the angular rate of the object and the resultant moment from robots and frictional torque will converge to zero, as shown in the equations below, respectively.

$$\lim_{t \rightarrow \infty} \omega(t) = 0. \quad (18)$$

$$\lim_{t \rightarrow \infty} \left(\sum_{i=1}^n (f_i(t)d_i) + M_f \right) = 0 \quad (19)$$

Proof 1:

From eqs. 2, 6, and ??, when the object rotates, we derive:

$$I\dot{\omega} + \mu_{0,r}I\frac{\omega}{\|\omega\|} + \mu_{1,r}\omega - \sum_{i=1}^n (f_i d_i) = 0. \quad (20)$$

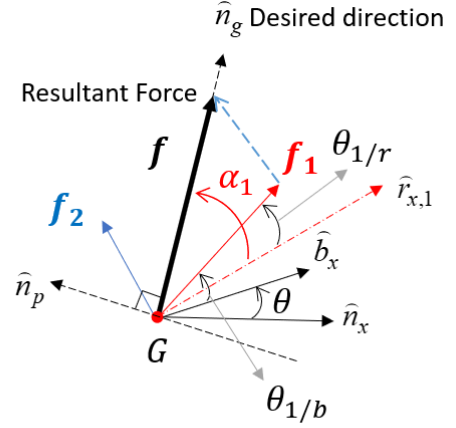


Fig. 7: Force vectors and coordinate frames of two pushing robots translated to the object's center of mass, enabling the analysis of rigid body motion with the corresponding denotations.

In *Scenario 1*, where there is no moment from robots $\sum_{i=1}^n (f_i d_i) = 0$. We have:

$$\dot{\omega} + a(\omega)\omega = 0, \quad (21)$$

where $a(\omega) = \mu_{0,r} \frac{1}{\|\omega\|} + \frac{\mu_{1,r}}{I} > 0 \forall \omega \neq 0$.

Denote the Lyapunov function:

$$V(t) = \frac{\omega(t)^2}{2} > 0 \forall \omega \neq 0. \quad (22)$$

$$\dot{V}(t) = \omega\dot{\omega} = -a(\omega)\omega^2 < 0 \forall \omega \neq 0. \quad (23)$$

From eqs. 22 and 23, Theorem 1 for Scenario 1 is proven.

In *Scenario 2*, we firstly investigate the case of two robots. Without loss of generality, we assume that robot 1 represents the *Right group* and robot 2 represents the *Left group*, having the lever arms satisfying conditons in eq. 14 as below:

$$d_1 = d_1^L < 0, d_2 = d_1^R > 0. \quad (24)$$

$$\alpha_1 < 0, \alpha_2 > 0. \quad (25)$$

The parameters f_i^0 can be found through the approximate static friction in the previous phase:

$$f_0^L = f_0^R = f_0. \quad (26)$$

As shown in Fig. 7, the relative orientation between the desired direction and the pushing force direction from robot i can be represented as:

$$\theta_{\widehat{r_{f,i}}, \widehat{n_g}} = \alpha_{i/r} - \theta_{i/r} = \theta_{\widehat{r_{f,i}}, \widehat{b_x}} + \theta_{\widehat{b_x}, \widehat{n_x}} - \theta_{\widehat{n_g}, \widehat{n_x}} \quad (27)$$

To simplify, we denote $\theta_{\widehat{r_{f,i}}, \widehat{b_x}} = \theta_{i/b}$, where $\theta_{i/b}$ is the angle of the pushing force of robot i in the body frame b , $\theta_{\widehat{b_x}, \widehat{n_x}} = \theta$ is the relative angle between the body and world

frames, and $\theta_{\widehat{n}_g, \widehat{n}_x} = \theta_g$ is the angle of the desired direction in the world frame. From there, we have:

$$\alpha_1 = \theta_{1/b} + \theta - \theta_g, \alpha_2 = \theta_{2/b} + \theta - \theta_g \quad (28)$$

If the desired direction changes relatively slowly compared to the object orientation, taking the derivative from the above equations, we have:

$$\dot{\alpha}_1 = \dot{\alpha}_2 = \dot{\theta} \quad (29)$$

From eqs. 15, 16, 25, 26, 28 and 29, we represent the resultant moment from robots as:

$$\sum_{i=1}^n (f_i d_i) = f_0 k_0 \mathbf{h}(\theta) + f_0 k_1 \dot{\theta} (d_1 - d_2), \quad (30)$$

where:

$$h(\theta) = \frac{d_1}{\sin(\theta_g - \theta_{1/b} - \theta)} + \frac{d_2}{\sin(\theta_{2/b} + \theta - \theta_g)} \quad (31)$$

$$\frac{dh(\theta)}{d\theta} = \frac{d_1 \cos(\theta_g - \theta_{1/b} - \theta)}{\sin^2(\theta_g - \theta_{1/b} - \theta)} - \frac{d_2 \cos(\theta_{2/b} + \theta - \theta_g)}{\sin^2(\theta_{2/b} + \theta - \theta_g)} < 0 \quad (32)$$

When θ ranges from $\theta_g - \theta_{2/b}$ to $\theta_g - \theta_{1/b}$, the value of $\mathbf{h}(\theta)$ ranges from $+\infty$ to $-\infty$. Combining with the condition as in eq. 32, there exists a unique solution $\theta = \theta_s$ such that $\mathbf{h}(\theta_s) = 0$. From eq. 31, we have:

$$d_1 \sin(\theta_{2/b} + \theta_s - \theta_g) + d_2 \sin(\theta_g - \theta_{1/b} - \theta_s) = 0 \quad (33)$$

In the case of multiple robots, there exists at least one solution: $\theta = \theta_s \in [\max(\theta_g - \theta_j^R), \min(\theta_g - \theta_k^L)]$ $\mathbf{h}(\theta_s) = 0$.

Define the system error based on the solution:

$$e(t) = \theta(t) - \theta_s, \quad (34)$$

$$\dot{e}(t) = \dot{\theta}(t) = \omega(t), \quad (35)$$

$$\ddot{e}(t) = \ddot{\theta}(t) = \dot{\omega}(t). \quad (36)$$

We aim to prove that the object angular rate $\omega(t) = \dot{e}(t)$ converges to 0 over time.

In case of two robots, replace $\theta(t) = e(t) + \theta_s$ to eq. 31 with the note $\mathbf{h}(\theta_s) = 0$, we derive:

$$h_e(e) = \sin(e)g(e) \quad (37)$$

where

$$g(e) = \frac{d_1 \cos(\theta_{2/b} + \theta_s - \theta_g) - d_2 \cos(\theta_g - \theta_{1/b} - \theta_s)}{\sin(|\alpha_1(e)|) \sin(|\alpha_2(e)|)} \quad (38)$$

Since the robots could only be in Left Group or Right Group, we have:

$$0 < \theta_{2/b} - \theta_{1/b} < \pi \quad (39)$$

$$\Leftrightarrow \sin((\theta_{2/b} + \theta_s - \theta_g) + (\theta_g - \theta_{1/b} - \theta_s)) > 0 \quad (40)$$

$$\Leftrightarrow \sin(\theta_{2/b} - \theta_s - \theta_g) \cos(\theta_g - \theta_{1/b} - \theta_s) + \sin(\theta_g - \theta_{1/b} - \theta_s) \cos(\theta_{2/b} - \theta_s - \theta_g) > 0 \quad (41)$$

Combining the above equation with eqs. 10, 11, 24, 33, and 42 we conclude:

$$g(e) < 0 \quad (42)$$

From eqs. 20, 35, 36 and 37, we derive a second-order system:

$$\ddot{e}(t) + b(\dot{e}(t)) + c(e(t)) = 0, \quad (43)$$

where

$$b(\dot{e}(t)) = \frac{1}{I} \left(\frac{\mu_{0,r}}{\|\dot{e}(t)\|} + \mu_{1,r} + f_0 k_1 (d_2 - d_1) \right) \dot{e}(t), \quad (44)$$

and

$$c(e(t)) = -\frac{f_0 k_0}{I} h_e(e(t)) = -\frac{f_0 k_0}{I} g(e(t)) \sin(e(t)). \quad (45)$$

From here, we can see that:

$$\dot{e}b(\dot{e}(t)) = \frac{1}{I} \left(\frac{\mu_{0,r}}{\|\dot{e}(t)\|} + \mu_{1,r} + f_0 k_1 (d_2 - d_1) \right) (\dot{e}(t))^2 \quad (46)$$

$$\dot{e}b(\dot{e}(t)) > 0 \text{ for } \dot{e} \neq 0 \quad (47)$$

Moreover, it is easy to notice that when $\theta \in [\theta_g - \theta_{2/b}, \theta_g - \theta_{1/b}]$ and $\max(\theta_{2/b} - \theta_{1/b}) = \frac{2\pi}{3}$, regardless the robot positions in group left and right, we have:

$$e \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3} \right] \quad (48)$$

From eqs. 42, 45, and 48, we have:

$$ec(e) > 0 \text{ for } e \neq 0 \quad (49)$$

Lyapunov-based analysis: Equation (43) is a second-order system with b and c as continuous functions. Based on the Invariant Set theorem [17], if Equations 47 and 49 are satisfied, the error function e converges asymptotically to zero.

Proof 1.2 Denote the Lyapunov function:

$$V(t) = \frac{\dot{e}^2}{2} + \int_0^e c(y) dy > 0, \quad \forall \dot{e} \neq 0. \quad (50)$$

Taking the derivative of Eq. 50, then combining it with Eqs. 24, 43, and 45,

$$\begin{aligned} \dot{V}(t) &= \dot{e}\ddot{e} + \dot{e}c(e) = -\dot{e}b(\dot{e}) \\ &= -\frac{1}{I} \left(\frac{\mu_{0,r}}{|\dot{e}|} + \mu_{1,r} + f_0 k_1 (d_2 - d_1) \right) \dot{e}^2 < 0 \quad \forall \dot{e} \neq 0 \end{aligned} \quad (51)$$

In the global invariant set theorem [17], the integral $\int_0^e c(y) dy$ is unbounded as $|x| \rightarrow \infty$ to achieve globally asymptotically stable of the equilibrium point. In our case, the integral $\int_0^e c(y) dy$ is unbounded as $e \rightarrow \theta_g - \theta_{2/b} - \theta_s$ and $e \rightarrow \theta_g - \theta_{2/b} - \theta_s$. Therefore, our system's equilibrium point, where $e = 0$ and $\dot{e} = 0$, is asymptotically stable when $e_{initial} \in [\theta_g - \theta_{2/b} - \theta_s, \theta_g - \theta_{1/b} - \theta_s]$ and $\forall \dot{e}$.

Note 1. over time, when the object angular rate reaches zero $\omega = 0$ and the resultant moment of robots is smaller than the maximum static frictional torque, the object reaches stagnation in rotation. e is constant and may not equal zero, but the resultant moment from robots and static frictional torque is zero, still satisfying in eq. 19.

Note 2. if the desired direction suddenly change and not in between the pushing force directions: $e_{initial} \notin [\theta_g - \theta_{2/b} - \theta_s, \theta_g - \theta_{1/b} - \theta_s]$

Theorem 2. Object velocity direction converges to the force direction:

Under a frictional environment with $\mu_{0,t}\mu_{1,t} \neq 0$, if the total force direction aligns with the desired direction $\widehat{n}_g$ and the total force magnitude is larger than static friction as in eq. 3, the object velocity direction converges to the force direction. The velocity magnitude will reach constantly in case of $\mu_{1,t} > 0$.

Proof 2: Denote $\widehat{n}_p$ as the axis that is perpendicular to the goal direction $\widehat{n}_g$, achieved by rotating $\widehat{n}_g$ by $\pi/2$. With our proposed control law, the total force on this axis is:

$$\sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_p = - \sum_{j=1}^{n^R} f_j^R \sin(|\alpha_j^R|) + \sum_{k=1}^{n^L} f_k^L \sin(|\alpha_k^L|) \quad (52)$$

Replace eqs. 15, 16 and 17 to eq. 52, we derive:

$$\sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_p = -k_1 \omega \left(\sum_{j=1}^{n^R} f_{j0}^L \sin(|\alpha_j^R|) + \sum_{k=1}^{n^L} f_{k0}^R \sin(|\alpha_k^L|) \right) \quad (53)$$

From eq. 18, the result of Theorem 1, and eq. 53, we derive:

$$\lim_{t \rightarrow \infty} \sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_p = 0 \quad (54)$$

We project the second Newton Law into the $\widehat{n}_p$ axis:

$$m \ddot{\mathbf{p}} \cdot \widehat{n}_p = \sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_p + \mathbf{F}_f \cdot \widehat{n}_p \quad (55)$$

From eqs. 54 and 55, when $t \rightarrow \infty$,

$$\dot{\mathbf{v}}_{\mathbf{G}} \cdot \widehat{n}_p + \left(\frac{\mu_{1,t}}{m} + \frac{\mu_{0,t}}{\|\mathbf{v}_{\mathbf{G}}\|} \right) (\mathbf{v}_{\mathbf{G}} \cdot \widehat{n}_p) = 0$$

With the frictional environment $\mu_{0,t} + \mu_{1,t} > 0$, $\dot{\mathbf{v}}_{\mathbf{G}} \cdot \widehat{n}_p$ will converge to zero.

The total force on the goal direction $\widehat{n}_g$ is:

$$\sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_g = \sum_{j=1}^{n^R} f_j^R \cos(\alpha_j^R) + \sum_{k=1}^{n^L} f_k^L \cos(\alpha_k^L). \quad (56)$$

When $t \rightarrow \infty$, $\omega \rightarrow 0$ and α_i is constant, we have:

$$\sum_{i=1}^n \mathbf{f}_i \cdot \widehat{n}_g \rightarrow F_0 > 0, \quad (57)$$

where

$$\begin{aligned} F_0 &= k_0 \left(\sum_{j=1}^{n^R} f_j^R \cos(\alpha_j^R) + \sum_{k=1}^{n^L} f_k^L \cos(\alpha_k^L) \right) \\ &= k_0 \left(\sum_{j=1}^{n^R} \frac{f_{j0}^L}{|\tan(\alpha_j^R(t))|} + \sum_{k=1}^{n^L} \frac{f_{k0}^R}{|\tan(\alpha_k^L(t))|} \right) \end{aligned} \quad (58)$$

Given the conditions of α_i as in eqs. 10 and 11, we need to prove the minimum value of F_0 can still overcome the static

friction regardless different pushing directions in both groups θ_j^R and θ_k^L and relative orientation. From eq. 58, we have:

$$F_0 \geq k_0 \left(\frac{\sum_{j=1}^{n^R} f_{j0}^L}{|\tan(\alpha_{j,max}^R(t))|} + \frac{\sum_{k=1}^{n^L} f_{k0}^R}{|\tan(\alpha_{k,max}^L(t))|} \right) \quad (59)$$

where $\alpha_{j,max}^R = \theta + \theta_{j,max}^R - \theta_g$ and $\alpha_{k,max}^L = \theta_g - \theta - \theta_{k,min}^L$ represent the max relative orientation can be achieve. From eq. 17, we derive:

$$\sum_{j=1}^{n^R} f_{j0}^L \geq F_{crit}, \sum_{k=1}^{n^L} f_{k0}^R \geq F_{crit}. \quad (60)$$

With $\theta \in [\theta_g - \theta_{j,max}^R, \theta_g - \theta_{k,min}^L]$, the inequality $F_0(\theta)$ will reach the minimum value:

$$F_{0,min} = \frac{2k_0 F_{crit}}{\tan(\theta_0)} \quad (61)$$

where $\theta_0 = \frac{\theta_{j,max} - \theta_{k,min}}{2} = \frac{\pi}{3}$. With $k_0 > 1$, $F_{0,min} > F_{crit}$, the resultant of robot force overcome the static friction regardless of robots pushing positions $\theta_{i/b}$, relative orientation $\theta_{i/b}$, and object rotation θ .

From eqs. 1 and 57, we have:

$$\begin{aligned} \dot{\mathbf{v}}_{\mathbf{G}} \cdot \widehat{n}_g + \frac{\mu_{1,t}}{m} \mathbf{v}_{\mathbf{G}} \cdot \widehat{n}_g + \frac{\mu_{0,t}}{m} \sigma(\mathbf{v}_{\mathbf{G}} \cdot \widehat{n}_g) &\rightarrow \frac{F_0}{m}, \\ \mathbf{v}_{\mathbf{G}} \cdot \widehat{n}_g &\rightarrow \frac{F_0 - \mu_{0,t}}{\mu_{1,t}}. \end{aligned} \quad (62)$$

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